Data Sharing and Collusion on Hybrid Platforms

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Abstract

Many online retail platforms, such as Amazon and JD.com, have recently begun providing sellers with proprietary consumer data. In this paper, we investigate how the sharing of such data can incentivize seller collusion through the usage of pricing algorithms. We find that the effect of data sharing on collusion sustainability and profitability depends on the mode operated by the platform. When a platform acts as both a host and seller, sharing data leads to more collusive outcomes. However, when a platform only intermediates purchases, sharing data hinders collusion. Our results suggest that imposing a ban on data sharing may be counterproductive and harmful to consumer welfare.

Keywords: data sharing, collusion, platform, personalized pricing

JEL Codes: D21, D40, L11

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1 Introduction

Online platforms, such as Amazon and JD.com, serve as intermediaries between buyers and sellers. This "gatekeeper" position allows them to collect large amounts of consumer-level data through tracking cookies, registration information, loyalty programs, etc. Moreover, platforms have begun sharing this proprietary consumer data with their sellers. For example, Amazon shares buyer identification and transaction information with their third-party sellers (see Choe et al., 2024). JD.com has also created a service that provides third-party sellers with consumer-level information that includes credit history, location, search history, and other preferences[1]. Both such acts assist sellers in offering targeted coupons to buyers or setting different prices based on consumer characteristics.

The implications of personalized pricing on consumer and seller welfare have been widely discussed in recent years by both policymakers and researchers. For instance, OECD (2018) states that personalized pricing "is typically pro-competitive and often enhances consumer welfare," but may also be harmful "by potentially enabling the exploitation of consumers and creating a perception of unfairness." The ambiguity in results can be somewhat seen in Rhodes and Zhou (2024). They find that the effect of personalized pricing on consumer welfare depends on whether full market coverage is satisfied. Although the static effects of personalized pricing are well understood, the effect that sharing data has on collusion sustainability and consumer welfare has not been heavily explored.

Moreover, an increasing number of decisions are now being supported through machine learning and algorithms that leverage big data. The emergence of such advanced tools has posed difficulties to fair competition on these online marketplaces. Researchers have found that algorithms can lead to raising prices and even learn to autonomously price collude without any prior knowledge of the market environment. One notable example of price hikes through algorithms is Amazon's scrapped "Project Nessie" which resulted in a lawsuit filed by the FTC.[2] Pricing algorithms are also being more widely adopted as platforms can provide sellers with simple pricing tools. Amazon provides a repricing software that is designed to automate pricing for sellers.[3] Previous research has shown that the usage of rule-based pricing strategies on Amazon has become more prevalent (see Chen et al., 2016, and Wang et al., 2022). Thus, the introduction and increasing use of pricing algorithms poses a deep concern for competition regulators. For example, the German Monopolies Commission

[1] Hu et al. (2023) provides a more detailed description of how JD.com shares consumer information.

[2] For experimental evidence of algorithmic collusion, see Calvano et al. (2020) and Klein (2021). Note that there is also some empirical evidence that suggests pricing algorithms may benefit consumers, see Hanspach et al. (2024).

[3] See `https://aws.amazon.com/marketplace/pp/prodview-lsel7dks2e7uw`.

(2018) states that "the increasing use of pricing algorithms makes collusion-related consumer damages more likely in the future." In addition. the question of whether antitrust liability can be established when various decisions are made by humans versus machines has been widely discussed (see, eg., OECD, 2017).

Policymakers have recently expressed concern about how sharing proprietary consumer data potentially contributes to both price discrimination and, in turn, algorithmic collusion. For example, Amazon has begun sharing data with their third-party sellers that include buyer identification and transaction information (see Choe et al., 2024). JD.com has also created a service that provides third-party sellers with consumer-level information that includes credit history, location, search history, and other preferences[4]. Both such acts assist sellers in offering targeted coupons to buyers or setting distinct prices given consumer characteristics. The UK's Competition and Markets Authority (2018) touches upon this fact and warns that pricing algorithms "in combination with the growth of 'Big Data'...might lead to personalized pricing." But, will this influx of personalized pricing from data sharing have implications on algorithmic collusion? Some competition authorities believe that sharing data potentially leads to supra-competitive pricing. One example is the *Preventing Algorithmic Collusion Act* which was introduced by the U.S. Congress that prohibits the use of a pricing algorithm that incorporates or was trained using proprietary data.[5] Although the legislation provides a first-step solution towards preventing algorithmic collusion, do algorithms respond to an increase in consumer information as one would expect? That is, does sharing data affect how algorithms price on online marketplaces, and if so, does the pricing become more anti-competitive?

In this paper, we analyze how sharing data affects sustainability of tacit collusion between two horizontally differentiated sellers intermediated by a data-rich platform.[6] We consider two different modes operated by the platform. First, we consider a platform that runs in "marketplace mode" where the platform's role is providing access between buyers and sellers. We also consider when the platform runs in "dual mode" where the platform is fully integrated with a seller. Under dual mode, the platform acts as both host and seller, e.g. Amazon with their Amazon Basics products. This means the platform sets prices in competition with the other seller.

We develop an infinitely repeated game where, given the market structure of the platform, the stage game is as follows. In each time period, the data-rich platform sets an ad-valorem

[4]Hu et al. (2023) provides a more detailed description of how JD.com shares consumer information.

[5]See `https://www.congress.gov/bill/118th-congress/senate-bill/3686`.

[6]We treat the mode of the platform as exogenous. Practically, platforms may choose which mode to operate based on the market environment. However, we are primarily focused on potential data sharing implications instead.

fee and chooses whether to share its data with sellers. If sellers have access to data, they can set personalized prices for each consumer instead of a standard uniform price. Observing their prices, consumers make their purchasing decisions. We assume each strategic player follows grim-trigger strategies to investigate their incentives for collusion.

Our first main result explores collusive outcomes given the platform operates as a marketplace. In this scenario, sellers face no information asymmetry. Under competition, platforms extract profits from the sellers to the point that they are on the margin of staying in the marketplace regardless of data sharing policy. Without data, price collusion is less profitable for sellers. But as sellers must internalize some profit loss on their turf when setting a lower uniform price, sellers are less incentivized to deviate. That is, collusion is more likely when under marketplace mode. Collusive profitability, from a platform perspective, depends on how patient sellers are. Because collusion is less profitable for sellers, platforms must set lower fees to ensure collusion. Thus, platforms generally prefer sharing data if sellers are sufficiently patient, vice versa.

If the platform operates under dual mode, sharing data in fact increases collusive sustainability and profitability. Now, the data-rich platform competes with the seller and has an informational advantage under no data sharing. The same logic applies as before, but with a reduced effect. That is, even though collusion is less profitable for the seller under no data sharing versus full data sharing, the seller must set a lower uniform price when deviating and incur a loss from consumers originally in their turf. However, because the platform has an advantage under collusion, this loss is greatly reduced in comparison to when the platform was in marketplace mode. This is the driving force behind the sustainability result. When both players have access to consumer data, collusive profits are highest and the platform can enforce a higher fee under collusion. Thus, profitability for the platform is greater with data sharing. Table 1 summarizes the key findings in the paper.

As consumer welfare is lowest when sellers price collude and have access to data, our analysis shows that the effects of data sharing on collusive outcomes are not black and white. From a regulation standpoint, banning the usage of proprietary data for pricing algorithms may be counterproductive. Instead, regulation policies should depend on the structure of the platform or specific product markets.

The rest of this paper goes as follows. Section 2 reviews the related literature. Section 3 describes the model of the stage game and dynamic strategy of the players. Sections 4 and 5 analyze collusive outcomes under marketplace mode and dual mode, respectively. Section 6 ends with concluding remarks.

	Collusive Sustainability	**Collusive Profitability**
Marketplace Mode	*No Data Sharing*	*No Data Sharing − if sellers are less patient* *Data Sharing − if sellers are more patient*
Dual Mode	*Data Sharing*	*Data Sharing*

Table 1: Platform preference on which mode to operate is based on collusive sustainability or profitability.

2 Related literature

This paper contributes to the literature on competitive personalized pricing. Our stage game follows a framework similar to that of Thisse and Vives (1988) which has been widely adopted to study spatial price discrimination. They find that personalized pricing reduces firm profits and increases consumer surplus under competition. Other related papers that use this framework include Shaffer and Zhang (2002) and Montes et al. (2019). Shaffer and Zhang (2002) studies competition between firms with asymmetric targeting costs for personalized pricing, whereas Montes et al. (2019) examines how a data intermediary should distribute data to firms for the purpose of personalized pricing. Anderson et al. (2023) and Rhodes and Zhou (2024) also focus on competitive personalized pricing by using a general discrete-choice model. Specifically, Anderson et al. (2023) allows consumers to opt-in to discounts which is costly to firms, while Rhodes and Zhou (2024) show that the welfare consequences discussed in Thisse and Vives (1988) can be reversed if the market is not fully covered.

We also touch on the viability of data sharing among competing sellers. Gradwohl and Tennenholtz (2023) find that selling only some consumers' data may lead to win-win outcomes for competing sellers. Both Hu et al. (2023) and Navarra et al. (2024) model a dual-role platform that can share data with its seller that it hosts. In particular, Hu et al. (2023) find that information sharing between the platform and seller should be regulated based on commission rate. Navarra et al. (2024) show that platforms are always incnetivized to share data, albeit they consider other data sharing scenarios than just full data sharing. We instead find that platforms unambiguously benefit by data sharing only under dual mode. Our analysis mainly differs in that we consider ad-valorem fees as opposed to transaction fees we are primarily focused on collusive outcomes under such information sharing policies.

At its core, our paper contributes to the literature that examines the interplay between collusion and price discrimination. Specifically, previous literature has shown that the effect price discrimination has on collusive behavior is ambiguous. For example, Helfrich and

Herweg (2016) use a linear city model and finds that a ban on third-degree price discrimination increases collusion sustainability. Döpper and Rasch (2024) similarly shows that a ban on second-degree price discrimination facilitates collusive outcomes. However, Peiseler et al. (2022) analyzes a setting where firms can third-degree price discriminate under private information. The addition of noisy signals on consumer preferences yields the opposite outcome. That is, they show that when signals are sufficiently noisy, a ban on third-degree price discrimination may reduce collusion. To the best of our knowledge, we are the first to bridge personalized pricing using a Hotelling framework and collusion.

Moreover, we address questions brought up in the fierce debate on algorithmic collusion. Many seminal works have discussed how pricing algorithms may autonomously collude. For example, Calvano et al. (2020) and Klein (2021) simulate competing reinforcement-learning algorithms and find that they can converge to collusive outcomes. In contrast, Miklós-Thal and Tucker (2019) suggest that better algorithms and demand forecasting potentially leads to lower prices and higher consumer welfare. Johnson et al. (2023) tackles the question from a platform design perspective. They show that a platform may enforce steering rules that can combat algorithmic collusion to the benefit of both consumers and the platform.

3 Description of the model

Consider the following stage game that represents a specific product market on a platform during a given time period.

A platform A operates in marketplace mode and hosts two competing sellers B_1 and B_2 along a unit interval [0,1] following Hotelling (1929). Sellers B_1 and B_2 are positioned at the beginning and end of the Hotelling line, respectively (i.e. $l_{B_1} = 0$ and $l_{B_2} = 1$). A measure one of consumers are uniformly distributed on the Hotelling line. They have at most one unit of demand, value both sellers' good at 1, and have an outside utility of 0. Also, consumers face a transportation cost τ per unit traveled where $0 < \tau < 1/2$.[7] We interpret τ as the degree of product differentiation. To summarize, a consumer located at $x \in [0, 1]$ purchasing a good from $i \in \{B_1, B_2\}$ gains the following utility:

$$U(x, p_i) = 1 - p_i - \tau|l_i - x|$$

where p_i is the price set by i.

Both sellers produce the good at marginal cost of 0 and pay an ad valorem fee $r \in [0, 1]$ for

[7]The restrictions on τ guarantee there will be full market coverage if a seller prices as a monopolist. This will simplify the analysis when solving the stage game for a platform operating in dual mode.

each good they sell to the platform. We assume that the platform is data-rich in the sense that it knows all consumer locations along the unit interval. Moreover, platform A has the option to share this consumer-level data with the sellers via its data-sharing policy $\mathcal{D} \in \{S, N\}$. If $\mathcal{D} = S$, the platform shares data on each consumer with the sellers which allows sellers to enforce personalized pricing (i.e. first-degree price discrimination). Otherwise if $\mathcal{D} = N$, the sellers sets a uniform price.

There are two crucial assumptions we require in the model. First, the seller has a reservation payoff of u where $0 < u < \frac{\tau}{8}$. This can be a result of the seller having a direct channel outside the platform or selling on some competing platform. The second key assumption is that the platform will always guarantee that the seller's profit is weakly greater than u by adjusting its ad valorem fee. This is a reduced-form way of saying that the platform sufficiently values the seller being on the platform. This can be attributed to consumers valuing having more purchasing choices. We provide a brief micro-foundation for this assumptions in Appendix B. Under these two assumptions, there will exist a pure strategy equilibrium even under asymmetric information, which to our knowledge, has not been explored in the related literature. Note that instead of this assumption, other papers require sequential pricing from competing sellers to ensure a pure strategy equilibrium. We avoid such an assumption, because once we consider a repeated game of this stage game, it would be ambiguous as to whether results are derived from the difference in timing structure or data sharing policy.

The timing of the game goes as follows:

1. A chooses $(r, \mathcal{D}) \in [0, 1] \times \{S, N\}$.

2. B_1 and B_2 observes $(r, \mathcal{D})$. If $\mathcal{D} = S$, B_1 and B_2 sets personalized prices $p_{B_1}(x)$ and $p_{B_2}(x)$, respectively, for all consumers. If $\mathcal{D} = N$, B_1 and B_2 sets uniform prices p_{B_1} and p_{B_2}, respectively. All pricing is done simultaneously.

3. Consumers observe prices and make their purchasing decisions.

3.1 Dynamic Strategy

To analyze how data sharing affects possible algorithmic collusion, we consider an infinitely repeated game of the stage game. We assume that all players are long-lived and share a common discount factor $\delta \in (0, 1)$ per period. Furthermore, under perfect monitoring, B_1 and B_2 follow grim trigger strategies *a la* Friedman (1971). That is, if any seller deviates from collusive pricing, the other seller prices according to the (static) Nash equilibrium of the stage game for each subsequent period. To summarize, the solution concept is SPNE in trigger strategies. We stick with grim-trigger strategies for the following reasons. First, in the context of Hotelling set ups, there is evidence that optimal punishment strategies tend to be similar to grim-trigger strategies (see Häckner (1996) and Liu and Serfes (2007)). Recent studies on algorithmic collusion have also shown that algorithms can learn sophisticated grim-trigger strategies to enforce supra-competitive prices. Finally, optimal punishment strategies come with the tradeoff of less tractable models. We will provide more specific details on how we characterize collusive sustainability in Sections 4 and 5.

4 Equilibrium in the non-integrated case

We proceed by solving for the Subgame Perfect Nash Equilibrium (SPNE) as well as collusive and deviation outcomes under data sharing (denoted with subscript S) and without data sharing (denoted with subscript N). Because sellers are symmetric and face the same fee, their profits under competition, collusion, and when deviating are the exact same.

4.1 Full Data Sharing, $\mathcal{D} = S$

Suppose A sets some fee r. The competitive pricing follows exactly from Thisse and Vives (1988). Seller B_1 sets price $\max\{\tau(1-2x), 0\}$ and seller B_2 sets price $\max\{\tau(2x-1), 0\}$ for a consumer located at x. As a result, seller payoffs can be characterized as

$$\pi_S^*(r) = (1-r)\int_0^{\frac{1}{2}} \tau(1-2x)dx = (1-r)\frac{\tau}{4}. \tag{1}$$

By backwards induction, platform A will set r to extract seller profit up to u. Specifically, A's optimal ad valorem fee r_S^* must satisfy $\pi_S^*(r_S^*) = u \iff r_S^* = 1 - \frac{4u}{\tau}$. Thus, under competition, A gains $\pi_{A,S} = 2\pi_S^*(r_S^*)\frac{r_S^*}{1-r_S^*} = \frac{\tau}{2} - 2u$ and both sellers end with u in profit. Under data sharing, each seller has full knowledge of consumer locations and tries to win over consumers on its rival's side with low prices. This induces sellers to charge low prices even for consumers who have strongly prefer their product. Naturally, the platform will then calibrate their fee to extract as much profit as possible.

Under full collusion, sellers can maximize industry profits by splitting the market and setting their price to each consumer's willingness to pay. This occurs with the following price schedule

$$p_S^c(x) = \begin{cases} 1 - \tau x & \text{if } x \leq \frac{1}{2} \\ 1 - (1-x)\tau & \text{if } x \geq \frac{1}{2} \end{cases}. \tag{2}$$

That is, each seller would attain a collusive profit of

$$\pi_S^c(r) = (1-r)\int_0^{\frac{1}{2}}(1-\tau x)dx = (1-r)(\frac{1}{2} - \frac{\tau}{8}). \tag{3}$$

Now given that a seller sets $p_S^c(x)$, the opposing seller's optimal deviating strategy would be to undercut $p_S^c(x)$ for all consumers. B_1's deviating price would then be $1-\tau x$ and B_2's deviating price would then be $1-\tau(1-x)$ for consumer located at x. Thus, deviating profit under full data sharing is

$$\pi_S^d(r) = (1-r)\int_0^{1}(1-\tau x)dx = (1-r)(1 - \frac{\tau}{2}). \tag{4}$$

4.2 No Data Sharing, $\mathcal{D} = N$

Again, suppose A sets some fee r. Similarly to the full data sharing scenario, the competitive payoffs follow closely from Thisse and Vives (1988) where each seller sets uniform price τ and obtains

$$\pi_N^*(r) = (1-r)\frac{\tau}{2}. \tag{5}$$

Using the same logic from before, A's optimal fee r_N^* is such that $\pi_N^*(r_N^*) = u \iff r_N^* = 1 - \frac{2u}{\tau}$. Thus, A will attain $\pi_{A,N} = 2\pi_N^*(r_N^*)\frac{r_N^*}{1-r_N^*} = \tau - 2u$ and both sellers are left with u profit. Here, we get the Thisse and Vives (1988) result where personalized pricing benefits every consumer. But, this also implies that the platform prefers to restrict data sharing in competition as their fee is dependent on seller pricing, i.e. $r_N^* > r_S^*$.

Because sellers are constrained to setting uniform prices, both sellers will set $p_N^c = 1 - \frac{\tau}{2}$ under full collusion. Like Full Data Sharing, sellers will split the market and each consumer purchases from the seller that is closer to them. However, as the sellers have no knowledge of individual locations, they cannot fully extract the consumer surplus. If a seller deviates, they would set their uniform price at $p_N^d = 1 - \tau$ to capture the entire market. Note that p_N^d represents the price set by a seller if they act as a monopolist. Thus, the collusive and deviating profits under no data sharing can be characterized as follows:

$$\pi_N^c(r) = (1-r)(\frac{1}{2} - \frac{\tau}{4}) \text{ and } \pi_N^d(r) = (1-r)(1-\tau).$$

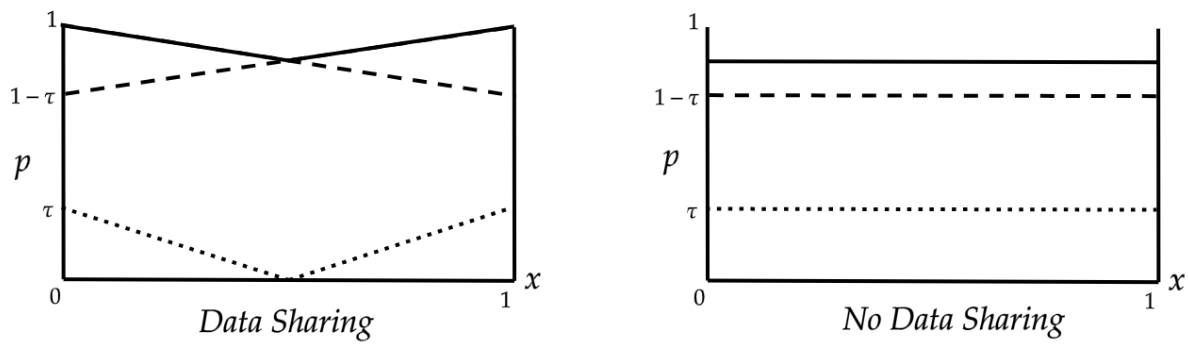

Figure 1: A graphical comparison of the price schedules based on different pricing policies under marketplace mode. Solid lines represent collusive prices, dashed lines represent deviating prices, and dotted lines represent competitive prices.

4.3 Comparison of Full Data Sharing vs. No Data Sharing

Given the presented dynamic environment, we say collusion is sustainable under policy $\mathcal{D} \in \{S, N\}$ as an SPNE if and only if $\frac{\pi^c_{\mathcal{D}}(r)}{1-\delta} \geq \pi^d_{\mathcal{D}}(r) + \frac{\delta}{1-\delta}\pi^*_{\mathcal{D}}(r^*_{\mathcal{D}})$ and $2\pi^c_{\mathcal{D}}(r)\frac{r}{1-r} \geq \pi_{A,\mathcal{D}}$ where r is the ad valorem fee set under collusive pricing for policy $\mathcal{D}$. This means δ must satisfy

$$\delta \geq \delta^{mkt}_{\mathcal{D}}(r) \equiv \frac{\pi^d_{\mathcal{D}}(r) - \pi^c_{\mathcal{D}}(r)}{\pi^d_{\mathcal{D}}(r) - \pi^*_{\mathcal{D}}(r^*_{\mathcal{D}})} \text{ for } \mathcal{D} \in \{S, N\}.$$

Here, $\delta^{mkt}_{\mathcal{D}}(r)$ represents the critical discount factor under policy k and measures collusive sustainability. A higher $\delta^{mkt}_{\mathcal{D}}(r)$ implies that the set of discount factors that can support collusion is smaller. Now, we are ready to compare our two data sharing policies to determine which is more sustainable for collusive pricing.

Lemma 1 *Given δ, the platform's optimal ad valorem fee to sustain collusion under marketplace mode is:*

$$r^{mkt}_{\mathcal{D}}(\delta) = \begin{cases} \max\{1 - \frac{\delta u}{\delta(1-\frac{\tau}{2})-\frac{1}{2}+\frac{3\tau}{8}}, 0\} & \textit{if } \mathcal{D} = S \\ \max\{1 - \frac{\delta u}{\delta(1-\tau)-\frac{1}{2}+\frac{3\tau}{4}}, 0\} & \textit{if } \mathcal{D} = N \end{cases}.$$

Lemma 1 characterizes the fee set by A to ensure collusion can be realistically sustained. For sufficiently low discount factors, collusion is not possible unless the fee is negative so we set it to 0. The properties of the fee are quite intuitive. Regardless of data sharing policy, the fee is weakly increasing in δ and decreasing in u. If sellers are more patient or have lower outside options, the platform can extract more seller profit by setting a higher fee under collusion. One last immediate observation is that for less patient sellers, $r^{mkt}_N \geq r^{mkt}_S$ and vice versa.

Proposition 1 *Under marketplace mode, collusion is more sustainable with no data sharing than with full data sharing.*

When the platform operates under marketplace mode, sellers are more willing to collude with no data sharing. The intuition is quite simple. Because the platform always extracts profits from the seller until they are at u profit in competition, collusion sustainability is essentially dependent on the magnitude of deviating profits compared to collusive profits. Although the collusive profits increase with data sharing as sellers can set personalized prices for each consumer, the deviating profits increase even more as each seller can capture the entire market by setting price to every consumers' willingness-to-pay. Instead, if a seller deviates under no data sharing, they must also lower prices for consumers they would have won under collusion. Thus, deviating is less lucrative under uniform pricing. So far, our analysis has only considered how collusive outcomes given some exogenous data sharing policy. But how will endogenizing $\mathcal{D}$ affect A's decision?

Corollary 1 *With endogenous $\mathcal{D}$, the following summarizes A's choice of data sharing in marketplace mode:*

- *If $\delta < \underline{\delta}_N$, A does not share data and there will be no collusion;*
- *If $\underline{\delta}_N \leq \delta < \underline{\delta}_S$, A does not share data and there will be collusion;*
- *If $\underline{\delta}_S \leq \delta$, A shares data and there will be collusion*

where $\underline{\delta}_N = \frac{1-\frac{3\tau}{2}+2u}{2(1-\tau+u)}$ *and* $\underline{\delta}_S$ *is such that* $\frac{r_S^{mkt}(\underline{\delta}_S)}{r_N^{mkt}(\underline{\delta}_S)} = \frac{4-2\tau}{4-\tau}$.

As mentioned earlier, the platform prefers not to share data in a competitive market. Accounting for this fact, our results from Proposition 1 still do not change. Indeed, collusion is more sustainable with no data sharing even when endogenizing $\mathcal{D}$. In a cartelized market, the platform will share data if sellers are patient enough. Recall from the discussion of Lemma 1 that the fee with data sharing exceeds that without data sharing. Coupled with the fact that given the same fee collusive profits are higher under no data sharing for sellers (i.e. $\pi_S^c(r) > \pi_N^c(r)$), as δ increases the platform will eventually prefer sharing data.

5 Equilibrium in the integrated case

In this section, we alter the market structure of the platform in the stage game. Platform A is now fully integrated with seller B_1, i.e. A now operates in dual mode. To simplify notation, we now denote seller B_2 as just B. The timing and all other assumptions of the game remain the same.

We assume players follow the same solution concept as in the non-integrated case, i.e. SPNE in grim trigger strategies. Given that the sellers (A and B) are not symmetric as opposed to in Section 4, players will now follow different critical discount factors. We discuss how this slightly changes the analysis in Section 5.3.

5.1 Full Data Sharing, $\mathcal{D} = S$

If A shares all the data with B, both players can set personalized prices for all consumers. First, observe that since A and B compete for consumers through localized Bertrand competition, competing price offers are driven down to marginal cost. In equilibrium, A is willing to decrease prices up to its effective marginal cost $rp_B(x)$ and a consumer located at $x \in [0,1]$ is indifferent between A and B iff

$$1 - rp_B(x) - \tau x = 1 - (1-x)\tau - p_B(x).$$

Rearranging terms and accounting for the fact that $p_B(x) \leq 1 - (1-x)\tau$, the optimal price B sets in equilibrium is

$$p^*_{B,S}(x,r) = \begin{cases} 0 & \text{if } x < \frac{1}{2} \\ \frac{(2x-1)\tau}{1-r} & \text{if } \frac{1}{2} \leq x < x^d(r) \\ 1-(1-x)\tau & \text{otherwise} \end{cases} \tag{6}$$

where $x^d(r) \equiv \min\{\frac{1-r(1-\tau)}{\tau(1+r)}, 1\}$. Note that B sets price down to 0 for consumers who prefer A's product. This follows the conventional logic of Bertrand competition as marginal cost is 0. However, for consumers who prefer B's product, A and B will price compete until the price is lowered such that A is willing to forgo the demand. Indeed, at a sufficiently low price, A's gain from pricing lower is lower than the gain from just letting B win the consumer and obtaining a proportion of the profit. $x^d(r)$ represents the point in which B is starting to get price capped at $1-(1-x)\tau$. One important thing to note is that $x^d(r) \to \frac{1}{2}$ as $r \to 1$ and $x^d(r) \to 1$ as $r \to 0$. This is quite intuitive as if r goes to 1, A will never price compete for the consumer (that prefers B's product) if $p_B(x) = 1-(1-x)\tau$. Similarly, if r goes to 0, the environment is just like that of Thisse and Vives (1988). Without r, we can just interpret A as a competing seller, instead of a dual-role platform.

Now, consider the consumers who prefer A's good. Because B is willing to price at 0, A

will set prices such that a consumer is indifferent between A and B. This occurs iff

$$1 - p_A(x) - \tau x = 1 - (1-x)\tau.$$

Thus, the optimal price A sets in equilibrium is

$$p^*_{A,S}(x,r) = \begin{cases} (1-2x)\tau & \text{if } x < \frac{1}{2} \\ rp^*_{B,S}(x,r) & \text{otherwise} \end{cases}. \tag{7}$$

This yields the following competitive payoffs for A and B:

$$\begin{aligned} \pi^*_{A,S}(r) &= \int_0^{\frac{1}{2}} (1-2x)\tau dx + \frac{r}{1-r}\pi^*_{B,S}(r) \\ \pi^*_{B,S}(r) &= \int_{\frac{1}{2}}^{x^d(r)} (2x-1)\tau dx + (1-r)\int_{x^d(r)}^{1} (1-(1-x)\tau)dx. \end{aligned}$$

Lemma 2 *The three following properties hold: (i)* $\frac{d\pi^*_{A,S}(r)}{dr} > 0$*; (ii)* $\frac{d\pi^*_{B,S}(r)}{dr} \leq 0$*; (iii)* $\pi^*_{B,S}(r)$ *is weakly concave in* r*.*

Let R^*_S denote the largest ad valorem fee such that $\pi^*_{B,S}(R^*_S) = u$. Lemma 2 implies that R^*_S is the profit-maximizing fee for A as A's equilibrium profit is increasing in r and B's equilibrium profit is weakly decreasing in r.

Under full collusion, A and B maximize joint profits in the market. A and B achieve this by only attracting consumers who prefer their product by setting their price schedule to that of (2) and obtaining the collusive payoffs

$$\pi^c_{A,S}(r) = (1+r)(\frac{1}{2} - \frac{\tau}{8}) \text{ and } \pi^c_{B,S}(r) = (1-r)(\frac{1}{2} - \frac{\tau}{8}).$$

Given the collusive price schedules, B's deviating strategy would be to undercut A's price for all consumers. A's deviating price would be the same, but only when the undercut price is greater than $rp^c_S(x)$. To summarize, A and B's deviating prices are

$$p^d_{A,S}(x,r) = \begin{cases} 1-\tau x & \text{if } x \leq x^d(r) \\ 1-\tau x^d(r) & \text{otherwise} \end{cases} \text{ and } p^d_{B,S}(x) = 1 - \tau(1-x).$$

Deviating profits under full data sharing would then be

$$\pi^d_{A,S}(r) = \int_0^{x^d(r)} (1-\tau x)dx + r\int_{x^d(r)}^1 (1-\tau(1-x))dx$$
$$\pi^d_{B,S}(r) = (1-r)\int_0^1 (1-\tau(1-x))dx = (1-r)(1-\frac{\tau}{2}).$$

Notice that collusive profits and deviating profits for B are exactly that from the non-integrated case in Section 4.1.

5.2 No Data Sharing, $\mathcal{D} = N$

If A is not allowed to share any data with B, seller B sets a uniform price for all consumers and only A can set personalized prices. To understand how we can arrive at a pure strategy equilibrium, consider our environment but without the two crucial assumptions: (i) the seller has an outside option of $u > 0$ and (ii) the platform prefers to keep the seller on the platform over selling on its own. This is akin to standard environments presented in the relevant literature. As previously mentioned, A and B compete for consumers through localized Bertrand competition. Thus, given some price p_B set by B, A's optimal strategy is to set $p_A(x) \geq rp_B$ such that consumers are indifferent between $p_A(x)$ and p_B. Specifically, A's optimal price schedule would be

$$\tilde{p}^*_A(x, p_B, r) = \begin{cases} (1-2x)\tau + p_B & \text{if } x \leq \tilde{x}(r, p_B) \equiv \min\{\frac{1}{2} + \frac{(1-r)p_B}{2\tau}, 1\} \\ 1-\tau x & \text{otherwise} \end{cases}. \quad (8)$$

However, B has an incentive to deviate to $p_B - \epsilon$ for some small ϵ and capture $(p_B - \epsilon)\tilde{x}$ in profit. This process will constantly continue until p_B is driven down to 0, but then B now has an incentive to increase prices above 0 as A is giving up half the market.

Lemma 3 *There exists a pure strategy equilibrium when $\mathcal{D} = N$. The equilibrium ad valorem fee and prices for A and B are characterized as follows:*

$$R^*_N = 1 - \frac{u}{1-\tau},\ p^*_{A,N}(x) = 1-\tau x,\ p^*_{B,N} = 1-\tau.$$

The logic of Lemma 3 goes as follows. When products are sufficiently similar, i.e. $\tau < \frac{1}{2}$, B is always willing to lower prices to attract the entire market demand. At a certain point, A is no longer willing to price compete and lets B have the market. However, A will recalibrate its ad valorem fee such that B's profit will be exactly u. In other words, B prices like a monopolist and A will extract profits up to the point that B still stays on the platform.

With the added assumption, A is no longer willing to undercut B. Indeed, if A prefers to keep A on the platform and B has some positive outside option, there is an effective "stopping point" at which A gives up price competing. It is easy to see that A will set the highest r such that $\pi^*_{B,N}(r) = u$. More specifically, A will set $R^*_N = 1 - \frac{u}{1-\tau}$ which gives us $\pi^*_{A,N}(R^*_N) = 1 - \tau - u$ and $\pi^*_{B,N}(R^*_N) = u$. Lemma 3 also implies that the equilibrium profits for A and B, respectively, are

$$\pi^*_{A,N}(r) = r(1-\tau) \text{ and } \pi^*_{B,N}(r) = (1-r)(1-\tau).$$

Lemma 4 *Under no data sharing, collusive prices are as follows:*

$$p^c_{A,N}(x) = 1 - \tau x \textit{ and } p^c_{B,N} = 1 - \frac{\tau}{3}.$$

Consequently, the deviating prices are:

$$p^d_{A,N}(x,r) = \begin{cases} 1 + \frac{2\tau}{3} - 2\tau x & \textit{if } \frac{2}{3} < x \leq x^d_N(r) \equiv \min\{\frac{1-r}{2\tau} + \frac{2+r}{6}, 1\} \\ 1 - \tau x & \textit{otherwise} \end{cases} \quad \textit{and } p^d_{B,N} = p^*_{B,N} = 1 - \tau$$

where A only has an incentive to deviate if and only if $r < \frac{3-2\tau}{3-\tau}$.

Under full collusion, A and B maximize industry profits just like before. Because B is restricted to setting a uniform price, B cannot fully extract consumer surplus on their end, so A is incentivized to attract more consumers under no data sharing. However, at some point, consumers are too far to reach for A, so they prefer to let those consumers purchase from B and extract profits through their ad valorem fee. Indeed, decreasing the amount of information B has access to means that the market is less efficient in terms of assessing consumers' willingness-to-pay. Lemma 4 implies that the collusive payoffs for A and B are:

$$\pi^c_{A,N}(r) = \frac{2}{3} - \frac{2\tau}{9} + r(\frac{1}{3} - \frac{\tau}{9}), \text{ and } \pi^c_{B,N}(r) = (1-r)(\frac{1}{3} - \frac{\tau}{9}).$$

Naturally, B's deviating price is to just set its price as if it was a monopolist under full market coverage, i.e. $\tau < \frac{1}{2}$. A only has an incentive to deviate from the collusive outcome if the ad valorem fee is sufficiently low such that there is room to undercut B's uniform price.

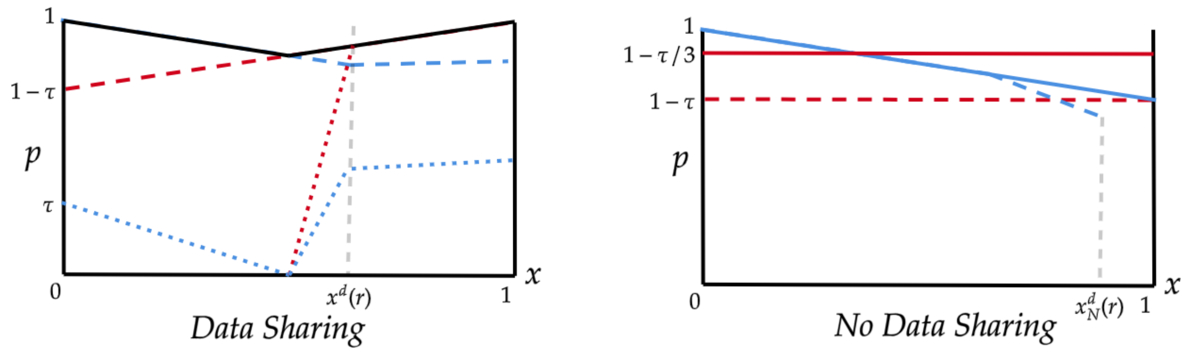

Figure 2: A graphical comparison of the price schedules based on different pricing policies under dual mode. Solid lines represent collusive prices, dashed lines represent deviating prices, and dotted lines represent competitive prices. Red lines represent the seller and blue lines represent the platform.

From Lemma 2, A and B's deviation payoffs are

$$\pi^d_{A,N}(r) = \begin{cases} \pi^c_{A,N}(r) & \text{if } r \geq \frac{3-2\tau}{3-\tau} \\ \frac{2}{3} - \frac{2\tau}{9} + \int_{\frac{2}{3}}^{x^d_N(r)} (1 + \frac{2\tau}{3} - 2\tau x)dx + r(1-\frac{\tau}{3})(1 - x^d_N(r)) & \text{otherwise} \end{cases},$$

$$\pi^d_{B,N}(r) = \pi^*_{B,N}(r) = (1-r)(1-\tau).$$

Notice that even though $\pi^d_{B,N}(r) = \pi^*_{B,N}(r)$, the profits will end up being different. As will be discussed later, the collusive ad valorem fee will be lower than that of the static Nash fee.

5.3 Comparison of Full Data Sharing vs. No Data Sharing

We say collusion is sustainable for player $j \in \{A, B\}$ under policy $\mathcal{D} \in \{S, N\}$ as an SPNE if and only if $\frac{\pi^c_{j,\mathcal{D}}(r)}{1-\delta} \geq \pi^d_{j,\mathcal{D}}(r) + \frac{\delta}{1-\delta}\pi^*_{j,\mathcal{D}}(r^*_{\mathcal{D}})$ where r is the ad valorem fee set under collusive pricing for policy k. Let $\delta_{j,\mathcal{D}}(r)$ be the critical discount factor for player $j \in \{A, B\}$ under policy $\mathcal{D} \in \{S, N\}$ where

$$\delta^{dual}_{j,\mathcal{D}}(r) \equiv \frac{\pi^d_{j,\mathcal{D}}(r) - \pi^c_{j,\mathcal{D}}(r)}{\pi^d_{j,\mathcal{D}}(r) - \pi^*_{j,\mathcal{D}}(R^*_{\mathcal{D}})}. \tag{9}$$

Naturally, collusion is sustainable under policy k if it is sustainable for both players. In other words, δ must satisfy

$$\delta \geq \delta^{dual}_{\mathcal{D}}(r) \equiv \max_{j \in \{A,B\}} \delta^{dual}_{\mathcal{D}}(r) \text{ for } \mathcal{D} \in \{S, N\}. \tag{CR}$$

Like in the non-integrated case, $\delta_{\mathcal{D}}^{dual}(r)$ follows the same interpretation as a measure for collusive sustainability.

We first characterize the platform's maximization problem to ensure collusion is sustained given δ and $\mathcal{D} \in \{S, N\}$:

$$\max_r \pi_{A,\mathcal{D}}^c(r) \text{ s.t. } (*) \begin{cases} \pi_{A,\mathcal{D}}^c(r) \geq \pi_{A,\mathcal{D}}^*(r) & (IR_A) \\ \pi_{B,\mathcal{D}}^c(r) \geq u & (IR_B) \\ \delta \geq \delta_{A,\mathcal{D}}^{dual}(r) & (CR_A) \\ \delta \geq \delta_{B,\mathcal{D}}^{dual}(r) & (CR_B) \end{cases}.$$

The CR_A and CR_B constraints are merely derived from condition (CR). Because $\pi_{A,\mathcal{D}}^c(r)$ is increasing in r, we just need to find the highest r such that $(*)$ is satisfied. Solving for both policies gives us the following Lemma.

Lemma 5 *Given δ, the platform's optimal ad valorem fee to sustain collusion under dual mode is:*

$$r_{\mathcal{D}}^{dual}(\delta) = \begin{cases} \max\{1 - \frac{\delta u}{(1-\frac{\tau}{2})\delta + \frac{3\tau}{8} - \frac{1}{2}}, 0\} & \text{if } \mathcal{D} = S \\ \max\{1 - \frac{\delta u}{(1-\tau)\delta + \frac{8\tau}{9} - \frac{2}{3}}, 0\} & \text{if } \mathcal{D} = N \end{cases}. \tag{10}$$

Lemma 5 gives us A's optimal fee to ensure collusion can be sustained under dual mode. Observe that $r_S^{dual}(\delta) = r_S^{mkt}(\delta)$ as B's collusive, deviating, and competitive payoff functions are the exact same under marketplace mode. Under no data sharing, we instead find that $r_N^{dual}(\delta) \leq r_N^{mkt}(\delta)$. The increase in fee under dual mode comes from the information asymmetry between A and B. Under dual mode, A captures more of the market demand leaving B with less profit when colluding. Because B's competitive payoff remains at u regardless of mode and B's collusive payoff is greatly reduced (compared to its deviating payoff), B is less incentivized to collude now. A must then reduce its collusive ad valorem fee to ensure B cooperates.

Proposition 2 *Under dual mode, collusion is more sustainable with full data sharing than with no data sharing.*

The result in Proposition 2 is primarily driven by the seller B's incentives for collusion. Because B's off-path payoff of u remains the same regardless of data sharing, we can focus our attention to the collusive and deviating payoffs for both policies. As discussed in Lemma 5, the seller's critical discount factor given r is the exact same under data sharing regardless of mode. Now, recall that when the platform operated under marketplace mode with no data sharing, sellers had to internalize some profit loss from consumers on their turf to deviate from collusive pricing and capture the entire market. This same effect remains under dual mode, but is severely diminished as A now captures more of the market demand when fully colluding due to its informational advantage. Additionally, due to a decrease in collusive profit for the seller, the platform must now enforce a lower ad valorem fee to facilitate collusion. This reduction in fee also makes collusion less profitable for the platform.

Also, note that we know from the proof of Proposition 2 that A's fee in competition is lower under data sharing, i.e. $R_S^* < R_N^*$. This is due to A's willingness to give up the market under no data sharing and instead set a higher fee to extract seller rent. In fact, for sufficiently low τ, A's competitive profit decreases under data sharing. Therefore, sharing data also may intensify off-path competition which leads to a lower ad-valorem fee and profit under competition for A which makes collusion more lucrative. Although for larger τ this does not hold, the aforementioned effect for the seller always holds and is the primary reason Proposition 2 holds.

Corollary 2 *Suppose $\delta \in (0, 1)$ such that collusion can be sustained for both players and A operates under dual mode. Then, collusion with full data sharing is more (less) profitable for A (B) than with no data sharing.*

Corollary 2 highlights an important point about sharing data under dual mode. The platform faces no draw backs as long as δ is sufficiently high. In contrast to marketplace mode where one policy might be more advantageous for collusion sustainability while the other may lead to higher collusive profits, collusion is both more sustainable and profitable under dual mode.

6 Concluding remarks

In this paper we analyzed the intersection of several emerging concerns for online platforms: increased sharing of private consumer information to sellers and the growing use of pricing algorithms which may lead to autonomous algorithmic collusion. We use a stylized model of seller competition in a repeated game to analyze this problem. We conclude that collusion is more likely and profitable for the platform when sharing data under dual mode. In contrast, collusion is less likely and platform profits are ambiguous when sellers have access to data in a standard marketplace.

Therefore, the key message that arises from our results is that a blanket ban on sharing data can potentially be harmful for consumers. Instead, one must assess the relationship between data sharing policy and platform structure on collusive outcomes. Naturally, this type of policy remedy requires a case-by-case analysis as opposed to a simple ban on data sharing to be effective.

The environment we examined is special. That is, our attention is restricted to two sellers and a market where total sales remain constant. Of course, many of the results rely on our critical assumptions in the stage game. However, these assumptions are required to keep pricing simultaneous. Otherwise, we cannot properly differentiate whether results are caused by a change in data sharing policy or timing structure. We conjecture that this is the primary reason why research bridging personalized pricing and collusion is less developed. Thus, Our paper faces the trade-off in being more stylized and providing a reliable take on data sharing on collusive outcomes.

Our model can be extended in many ways. First, we do not consider flexible data sharing where the platform can share subsets of consumer data along the line. This is explored in Navarra et al. (2024), but only in a static setting. Also, platforms share data with all sellers in our model. One could allow platforms to choose different data sharing decisions for each seller when operating in marketplace mode and analyze how it affects collusive outcomes. This comes with the caveat that there will not be a pure strategy equilibrium in the stage game, so additional assumptions would have to be imposed to guarantee a tractable and reliable model. Lastly, we assume that each set of consumers is unique in each time period. A possible extension could be to consider returning consumers in which sellers can learn about consumer locations along the Hotelling line.

References

Anderson, Simon, Alicia Baik, and Nathan Larson (2023) "Price Discrimination in the Information Age: Prices, Poaching, and Privacy with Personalized Targeted Discounts," *The Review of Economic Studies*, 90 (5), 2085–2115, 10.1093/restud/rdac073.

Calvano, Emilio, Giacomo Calzolari, Vincenzo Denicolò, and Sergio Pastorello (2020) "Artificial Intelligence, Algorithmic Pricing, and Collusion," *American Economic Review*, 110 (10), 3267–97, 10.1257/aer.20190623.

Chen, Le, Alan Mislove, and Christo Wilson (2016) "An Empirical Analysis of Algorithmic Pricing on Amazon Marketplace," in *Proceedings of the 25th International Conference on World Wide Web*, 1339–1349, 10.1145/2872427.2883089.

Choe, Chongwoo, Jiajia Cong, and Chengsi Wang (2024) "Softening Competition Through Unilateral Sharing of Customer Data," *Management Science*, 70 (1), 526–543, 10.1287/mnsc.2023.4689.

Competition and Markets Authority (2018) "Pricing Algorithms: Economic working paper on the use of algorithms to facilitate collusion and personalised pricing."

Döpper, Hendrik and Alexander Rasch (2024) "Combinable products, price discrimination, and collusion," *International Journal of Industrial Organization*, 94, 103071, https://doi.org/10.1016/j.ijindorg.2024.103071.

Friedman, James W. (1971) "A Non-cooperative Equilibrium for Supergames," *The Review of Economic Studies*, 38 (1), 1–12, 10.2307/2296617.

German Monopolies Commission (2018) "Algorithms and Collusion," *Biennial Report*, `https://www.monopolkommission.de/images/HG22/HGXXII_Summary.pdf`.

Gradwohl, Ronen and Moshe Tennenholtz (2023) "Selling Data to a Competitor," `https://arxiv.org/abs/2302.00285`.

Hanspach, Philip, Geza Sapi, and Marcel Wieting (2024) "Algorithms in the marketplace: An empirical analysis of automated pricing in e-commerce," *Information Economics and Policy*, 69, 101111, https://doi.org/10.1016/j.infoecopol.2024.101111.

Helfrich, Magdalena and Fabian Herweg (2016) "Fighting collusion by permitting price discrimination," *Economics Letters*, 145, 148–151, https://doi.org/10.1016/j.econlet.2016.05.024.

Hotelling, Harold (1929) "Stability in Competition," *The Economic Journal*, 39 (153), 41–57, http://www.jstor.org/stable/2224214.

Hu, Yihong, Guo Li, Mengqi Liu, and Shengnan Qu (2023) "Information Sharing and Personalized Pricing in Online Platforms," *Production and Operations Management*, 10591478231225178, 10.1177/10591478231225178.

Häckner, Jonas (1996) "Optimal symmetric punishments in a Bertrand differentiated products duopoly," *International Journal of Industrial Organization*, 14 (5), 611–630, https://doi.org/10.1016/0167-7187(95)00509-9.

Johnson, Justin P., Andrew Rhodes, and Matthijs Wildenbeest (2023) "Platform Design When Sellers Use Pricing Algorithms," *Econometrica*, 91 (5), 1841–1879, https://doi.org/10.3982/ECTA19978.

Klein, Timo (2021) "Autonomous algorithmic collusion: Q-learning under sequential pricing," *The RAND Journal of Economics*, 52 (3), 538–558, https://doi.org/10.1111/1756-2171.12383.

Liu, Qihong and Konstantinos Serfes (2007) "Market segmentation and collusive behavior," *International Journal of Industrial Organization*, 25 (2), 355–378, https://doi.org/10.1016/j.ijindorg.2006.05.004.

Miklós-Thal, Jeanine and Catherine Tucker (2019) "Collusion by Algorithm: Does Better Demand Prediction Facilitate Coordination Between Sellers?" *Management Science*, 65 (4), 1552–1561, 10.1287/mnsc.2019.3287.

Montes, Rodrigo, Wilfried Sand-Zantman, and Tommaso Valletti (2019) "The Value of Personal Information in Online Markets with Endogenous Privacy," *Management Science*, 65 (3), 1342–1362, 10.1287/mnsc.2017.2989.

Navarra, Federico, Flavio Pino, and Luca Sandrini (2024) "Mandated data-sharing in hybrid marketplaces," ZEW Discussion Papers 24-051, Mannheim, https://hdl.handle.net/10419/300680.

OECD (2017) "Algorithms and Collusion: Competition Policy in the Digital Age," *OECD Roundtables on Competition Policy Papers* (206), https://doi.org/10.1787/258dcb14-en.

——— (2018) "Personalised pricing in the digital era," *OECD Roundtables on Competition Policy Papers* (222), https://doi.org/10.1787/db4d9c9c-en.

Peiseler, Florian, Alexander Rasch, and Shiva Shekhar (2022) "Imperfect information, algorithmic price discrimination, and collusion," *The Scandinavian Journal of Economics*, 124 (2), 516–549, https://doi.org/10.1111/sjoe.12465.

Rhodes, Andrew and Jidong Zhou (2024) "Personalized Pricing and Competition," *American Economic Review*, 114 (7), 2141–70, 10.1257/aer.20221524.

S.3686 - 118th Congress (2024) "Preventing Algorithmic Collusion Act of 2024," `https://www.congress.gov/bill/118th-congress/senate-bill/3686`.

Shaffer, Greg and Z. John Zhang (2002) "Competitive One-to-One Promotions," *Management Science*, 48 (9), 1143–1160, `http://www.jstor.org/stable/822606`.

Thisse, Jacques-Francois and Xavier Vives (1988) "On The Strategic Choice of Spatial Price Policy," *The American Economic Review*, 78 (1), 122–137, `http://www.jstor.org/stable/1814702`.

Wang, Qiaochu, Yan Huang, and Param Vir Singh (2022) "Algorithms, Artificial Intelligence and Simple Rule Based Pricing," *SSRN Electronic Journal*, 10.2139/ssrn.4144905.

A Proofs.

Proof of Lemma 1.

Using the analysis in Section 4.1 and Section 4.2, we know

$$\delta_S^{mkt}(r) = \frac{(1-r)(\frac{1}{2}-\frac{\tau}{4})}{(1-r)(1-\frac{\tau}{2})-u} \text{ and } \delta_N^{mkt}(r) = \frac{(1-r)(\frac{1}{2}-\frac{3\tau}{4})}{(1-r)(1-\tau)-u}.$$

It can be easily verified that $\delta_{\mathcal{D}}^{mkt}(r)$ is strictly increasing within our parameter restrictions. For collusion to be sustainable under $\mathcal{D} \in \{S, N\}$, it must be that $\delta \geq \delta_{\mathcal{D}}^{mkt}(r)$ and as A's profit is strictly increasing in r, A's optimal ad valorem fee to sustain collusion is to set r such that $\delta = \delta_{\mathcal{D}}^{mkt}(r)$ which gives the presented result.

Proof of Proposition 1.

For collusion to be sustainable under N, it must be that $\delta \geq \delta_N^{mkt}(r)$ and $2\frac{r}{1-r}\pi_N^c(r) \geq \tau - 2u$. This occurs for

$$\frac{\tau - 2u}{1-\frac{\tau}{2}} \leq r \leq r_N^{mkt}(\delta).$$

Note that the RHS is increasing in δ under our parameter restrictions, so the lowest δ such that collusion is sustainable is when the LHS=RHS. Solving this gives the lowest δ such that collusion is sustainable under no data sharing as

$$\underline{\delta}_N = \frac{1-\frac{3\tau}{2}+2u}{2(1-\tau+u)} < \frac{1}{2} < \delta_S^{mkt}(0)$$

where $\delta_S^{mkt}(0)$ is a lower bound on how sustainable collusion is under full data sharing.

Proof of Corollary 1.

First, observe that the competitive payoff of A is higher under no data sharing. Therefore, the first item follows the proof of Proposition 1. The collusive profit for A is higher under S than N iff

$$2r_S^{mkt}(\delta)(\frac{1}{2}-\frac{\tau}{8}) > 2r_N^{mkt}(\delta)(\frac{1}{2}-\frac{\tau}{4}) \iff \frac{r_S^{mkt}(\delta)}{r_N^{mkt}(\delta)} > \frac{4-2\tau}{4-\tau}. \tag{A1}$$

Note that for $\underline{\delta}_N < \delta \leq \frac{\frac{1}{2}-\frac{3\tau}{8}}{1-\frac{\tau}{2}-u}$, $r_S^{mkt}(\delta) = 0$, so (A1) cannot be satisfied. It can be easily verified that for $\delta > \frac{\frac{1}{2}-\frac{3\tau}{8}}{1-\frac{\tau}{2}-u}$, $\frac{r_S^{mkt}(\delta)}{r_N^{mkt}(\delta)}$ is strictly increasing in δ and as $\delta \to 1$, $\frac{r_S^{mkt}(\delta)}{r_N^{mkt}(\delta)} > 1 > \frac{4-2\tau}{4-\tau}$. Thus, there must exist some $\underline{\delta}_S$ such that for all $\delta > \underline{\delta}_S$, (A1) is satisfied which gives us items 2 and 3.

Proof of Lemma 2.

Observe that if $r < 1-\tau$, $\pi^*_{B,S}(r) = \frac{\tau}{4}$. Otherwise,

$$\begin{aligned}
\pi^*_{B,S}(r) =& \int_{\frac{1}{2}}^{x^d(r)} (2x-1)\tau dx + (1-r)\int_{x^d(r)}^{1} (1-(1-x)\tau)dx \\
=& \tau[(x^d(r))^2 + \frac{1}{4} - x^d(r)] + (1-r)(1-\tau)(1-x^d(r)) + (1-r)\tau(\frac{1}{2} - \frac{(x^d(r))^2}{2}) \\
=& \tau[\frac{(x^d(r))^2}{2} - \frac{1}{4}] + (1-x^d(r))(1-r) + \frac{r\tau}{2}(1-x^d(r))^2 \\
=& \frac{(1-r+r\tau)^2}{2\tau(1+r)} + (1-r)\frac{\tau+r-1}{\tau(1+r)} + r\frac{\tau+2r-2-r\tau}{2(1+r)} - \frac{\tau}{4} < \frac{\tau}{4}.
\end{aligned}$$

It can be then shown that

$$\frac{d\pi^*_{B,S}(r)}{dr} = \begin{cases} 0 & \text{if } r < 1-\tau \\ \frac{3-4\tau-2r+\tau^2-r^2}{2\tau(1+r)^2} & \text{otherwise} \end{cases} \leq 0$$

which gives us item (ii). For $r < 1-\tau$, the proof of (i) is trivial. Solving for (i) when $r > 1-\tau$, we have

$$\begin{aligned}
\frac{d\pi^*_{A,S}(r)}{dr} = \frac{\pi^*_{B,S}(r)}{1-r} + \frac{r\pi^*_{B,S}(r)}{(1-r)^2} + \frac{r\pi^{*\prime}_{B,S}(r)}{1-r} > 0 &\iff \frac{\pi^*_{B,S}(r)}{1-r} + r\pi^{*\prime}_{B,S}(r) > 0 \\
&\iff (1-r)(\underbrace{4r-(1+r)(1+r^2)+2\tau(1-r)}_{\equiv z(r)}) + 2\tau^2 r > 0.
\end{aligned}$$

Note that a sufficient condition for the statement to hold is to show that $z(r) > 0$ which can be easily verified. When $r > 1-\tau$,

$$\frac{d^2\pi^*_{B,S}(r)}{dr^2} = \frac{-4+4\tau-\tau^2}{\tau(1+r)^3} < 0.$$

Therefore, it is immediate that $\pi^*_{B,S}(r)$ is weakly concave and (iii) holds.

Proof of Lemma 3.

Consider B's optimal price when selling by itself (as a monopolist). A consumer purchases from B iff

$$1-(1-x)\tau - p_B \geq 0 \iff x \geq 1 - \frac{1-p_B}{\tau}.$$

Thus, B's profit maximization problem is $\max_{p_B}\{p_B \min\{1, \frac{1-p_B}{\tau}\}\}$. This is a standard monopoly problem which yields the optimal price $p^*_B = 1-\tau$.

Suppose A sets r. We first derive A's optimal price schedule $\tilde{p}_A^*(x, p_B)$ given a price p_B. Given p_B, a consumer is indifferent between purchasing from A and B iff

$$1 - p_A(x) - \tau x = 1 - p_B - \tau(1-x) \iff p_A(x) = (1-2x)\tau + p_B.$$

A will only set such a price if $p_A(x) \geq rp_B \iff x \leq \tilde{x}(r, p_B) \equiv \min\{\frac{1}{2} + \frac{(1-r)p_B}{2\tau}, 1\}$. Otherwise, A is willing to give up the consumer to B. This gives us $\tilde{p}_A^*(x, p_B, r)$ from (8). Also, observe that A's deviation must still guarantee that B's profit is at least u. That is, it must be that

$$\pi_{B,N} = (1-r)p_B(1 - \tilde{x}'(r, p_B)) \geq u \iff \tilde{x}'(r, p_B) \leq 1 - \frac{u}{(1-r)p_B}$$

where $\tilde{x}'(r, p_B)$ represents the marginal consumer after A deviates. If $\tilde{x}(r, p_B) \leq \tilde{x}'(r, p_B)$, then $\tilde{p}_A^*(x, p_B^*)$ is A's optimal deviating price schedule. If $\tilde{x}(r, p_B) > \tilde{x}'$, then $\tilde{p}_A'^*(x, p_B^*)$ is A's optimal deviating price schedule where

$$\tilde{p}_A'^*(x, p_B^*) = \begin{cases} (1-2x)\tau + p_B & \text{if } x \leq \tilde{x}' \\ 1 - \tau x & \text{otherwise} \end{cases}.$$

Now, given A's optimal deviating price schedule, B has a profitable deviation from p_B^* to $p_B^* - \epsilon$ where ϵ is near 0 to earn $(1-r)(p_B^* - \epsilon)$ profit. We can continue this process of price adjustments until B adjusts its price to $\underline{p}_B$ such that $(1-r)\underline{p}_B = u > 0$. A has no incentive to deviate and readjust its price schedule. By backwards induction, A sets the highest r such that $\underline{p}_B = \frac{u}{1-r} \leq 1-\tau$. That is, the optimal fee $R_N^* = 1 - \frac{u}{1-\tau}$ which implies $\underline{p}_B = p_B^* = 1 - \tau$.

Proof of Lemma 4.

It is immediate that $p_{A,N}^c(x) = 1 - \tau x$ as consumer surplus is fully extracted for $x \leq \frac{1}{2}$. B sets a uniform price $p_B = 1 - \tau(1 - \bar{x})$ where $\bar{x}$ is the marginal consumer that is indifferent between $p_{A,N}^c(x)$ and p_B. The problem boils down to:

$$\max_{\bar{x}} \int_0^{\bar{x}} (1 - \tau x)dx + \int_{\bar{x}}^1 (1 - \tau(1 - \bar{x}))dx.$$

Solving gives that the optimal $\bar{x} = \frac{2}{3}$ which gives us $p_{B,N}^c = 1 - \frac{\tau}{3}$. Given that $p_{A,N}^c(x) = 1 - \tau x$, it is obvious that $p_{B,N}^d = 1 - \tau$, as when $\tau < \frac{1}{2}$, $1 - \tau$ is the price B sets as a monopolist (see proof of Lemma 3). For A's deviating price, notice that A will keep its price of $1 - \tau x$ for $x < \frac{2}{3}$. A can undercut consumers $x \geq \frac{2}{3}$ and set the highest price such that they are indifferent between buying from A and B which is satisfied iff

$$1 - p_A(x) - \tau x = 1 - (1-x)\tau - p_{B,N}^c \iff p_A(x) = 1 + \frac{2\tau}{3} - 2\tau x.$$

A is only incentivized to undercut B if

$$1 + \frac{2\tau}{3} - 2\tau x \geq r p^c_{B,N} \iff x \leq \frac{1-r}{2\tau} + \frac{2+r}{6}.$$

Thus, A is only incentivized to deviate if $\frac{1-r}{2\tau} + \frac{2+r}{6} > \frac{2}{3} \iff r < \frac{3-2\tau}{3-\tau}$.

Proof of Lemma 5.

Consider when $\mathcal{D} = N$. Fee r^{dual}_N satisfies the (IR_A) and (IR_B) constraints iff

$$1 - \frac{6\tau + 9u}{3-\tau} \leq r^{dual}_N \leq 1 - \frac{9u}{3-\tau}.$$

To satisfy (CR_B), it must be that

$$\delta \geq \delta^*_{B,N} = \frac{\pi^d_{B,N}(r^{dual}_N) - \pi^c_{B,N}(r^{dual}_N)}{\pi^d_{B,N}(r^{dual}_N) - \pi^*_{B,N}(R^*_N)} = \frac{(1-r^c_N)(\frac{2}{3} - \frac{8\tau}{9})}{(1-r^c_N)(1-\tau) - u} \iff r^{dual}_N \leq 1 - \frac{\delta u}{(1-\tau)\delta - \frac{2}{3} + \frac{8\tau}{9}}.$$

First, because $\delta^{dual}_{A,N}$ is weakly decreasing in r (see proof of Proposition 2), A will set $r^{dual}_N = \min\{1 - \frac{9u}{3-\tau}, 1 - \frac{\delta u}{(1-\tau)\delta - \frac{2}{3} + \frac{8\tau}{9}}\}$. Next, it is easy to verify that $r^{dual}_N = 1 - \frac{\delta u}{(1-\tau)\delta - \frac{2}{3} + \frac{8\tau}{9}}$ when $\delta \geq \frac{6-8\tau}{9-9\tau}$. When $\delta < \frac{6-8\tau}{9-9\tau}$, (CR_B) cannot be satisfied, so it is without loss to keep r^{dual}_N as $1 - \frac{\delta u}{(1-\tau)\delta - \frac{2}{3} + \frac{8\tau}{9}}$ as long as it is ≥ 0. For $\mathcal{D} = S$, we apply the same logic to arrive at $r^{dual}_S = \max\{1 - \frac{\delta u}{(1-\frac{\tau}{2})\delta + \frac{3\tau}{8} - \frac{1}{2}}, 0\}$. Note that this does not mean the fees necessarily ensure collusion. But when collusion can be sustained, these fees are optimal.

Proof of Proposition 2.

It is easy to verify that $\delta^{dual}_{A,N}(r)$ is continuous and differentiable for $r < \frac{3-2\tau}{3-\tau}$ and equal to 0 otherwise. For all other $j \in \{A, B\}$ and $k \in \{S, N\}$, $\delta^{dual}_{j,k}(r)$ is continuous and differentiable for $r \in (0, 1)$. When $\delta^{dual}_{j,k}$(r) is continuous and differential, observe that (9) implies

$$\delta^{dual\prime}_{j,k}(r) = \frac{\pi^{d\prime}_{j,k}(r) - \pi^{c\prime}_{j,k}(r)}{\pi^d_{j,k}(r) - \pi^*_{j,k}(R^*_k)} - \frac{\pi^d_{j,k}(r) - \pi^c_{j,k}(r)}{(\pi^d_{j,k}(r) - \pi^*_{j,k}(R^*_k))^2}\pi^{d\prime}_{j,k}(r) \leq (\geq)\, 0$$
$$\iff (1 - \delta^{dual}_{j,k}(r))\pi^{d\prime}_{j,k}(r) - \pi^{c\prime}_{j,k}(r) \leq (\geq)\, 0.$$

It is easy to verify that $\delta^{dual\prime}_{A,k}(r) \leq 0$ and $\delta^{dual\prime}_{B,k}(r) \geq 0$ given our parameter restrictions. When $r > \frac{3-2\tau}{3-\tau}$, $\delta^*_{A,N}(r) = 0$. Thus, $\delta^{dual}_{A,N}(r)$ is weakly decreasing. Moreover, it can be easily shown that $\delta^{dual}_{A,S}(r)$ is strictly decreasing and $\delta^{dual}_{B,S}(r)$ is strictly increasing for $r \in (0, 1)$.

Lemma A.1 *The optimal ad valorem fee in the static Nash equilibrium under full data sharing*

$R^*_S < 1-2u$ which implies

$$\delta^{dual}_{A,S}(r) < \tilde{\delta}^{dual}_{A,S}(r) \equiv \frac{\pi^d_{A,S}(r) - \pi^c_{A,S}(r)}{\pi^d_{A,S}(r) - \frac{\tau}{4} - (\frac{1}{2} - u)}.$$

Similarly, $\delta^{dual\prime}_{A,S}(r) < 0$.

Proof of Lemma A.1.

Evaluating $\pi^*_{B,S}(r)$ and $\frac{d\pi^*_{B,S}(r)}{dr}$ at $r = 1-2u$, we get

$$\begin{aligned}
\pi^*_{B,S}(1-2u) =& \frac{(2u+(1-2u)\tau)^2}{4\tau(1-u)} + \frac{u(\tau-2u)}{\tau(1-u)} + \tau(1-2u)\frac{(u\tau-2u)}{2\tau(1-u)} - \frac{\tau}{4} \\
=& \frac{4u\tau - 4u^2 + \tau^2 - 2u\tau^2}{4\tau(1-u)} - \frac{\tau}{4} < u \\
\iff& 4u\tau - 4u - \tau^2 < 0
\end{aligned}$$

which always holds. Because $\pi^*_{B,S}(r)$ is weakly decreasing in r and weakly concave, A must set $R^*_S < 1-2u$ to guarantee $\pi^*_{B,S}(r) \geq u$. The rest of the results in the lemma are immediate. ■

Lemma A.2 *The following two statements hold: (i) at $\tilde{r}$ such that $\delta^{dual}_{B,S}(\tilde{r}) = \delta^{dual}_{B,N}(0)$, $\delta^{dual}_{A,S}(\tilde{r}) < \delta^{dual}_{B,S}(\tilde{r})$ and (ii) $\min\{\delta^{dual}_{A,S}(0), 1\} > \delta^{dual}_{B,S}(0)$.*

Proof of Lemma A.2. For the first item, $\delta^{dual}_{B,S}(\tilde{r}) = \delta^{dual}_{B,N}(0) = \frac{6-8\tau}{9(1-\tau-u)}$ when $\tilde{r} = r^c_S\big|_{\delta=\delta^{dual}_{B,N}(0)}$. That is,

$$\tilde{r} = 1 - \frac{\delta u}{(1-\frac{\tau}{2})\delta + \frac{3\tau}{8} - \frac{1}{2}}\Bigg|_{\delta=\frac{6-8\tau}{9(1-\tau-u)}} = 1 - \frac{(6-8\tau)u}{\frac{3}{2} - \frac{25}{8}\tau + \frac{5}{8}\tau^2 + (\frac{9}{2} - \frac{27}{8}\tau)u} > 1-\tau.$$

When $r = 1-\tau$,

$$\begin{aligned}
&\tilde{\delta}^{dual}_{A,S}(r) = 1 - \frac{1-\tau+\frac{\tau^2}{8} - (1-u)}{1-\frac{3\tau}{4} - (\frac{1}{2}-u)} < \frac{6-8\tau}{9(1-\tau-u)} = 1 - \frac{3-\tau-9u}{9-9\tau-9u} \\
\iff& (3-\tau-9u)(\frac{1}{2} + u - \frac{3\tau}{4}) < (9-9\tau-9u)(u-\tau+\frac{\tau^2}{8}) \\
\iff& u < \underbrace{\frac{\frac{3}{2} + \frac{25}{4}\tau - \frac{75}{8}\tau^2 + \frac{9}{8}\tau^3}{\frac{21}{2} - \frac{23}{4}\tau - \frac{9}{8}\tau^2}}_{>\frac{\tau}{8} \text{ for } 0<\tau<\frac{1}{2}}
\end{aligned}$$

where $\tilde{\delta}^{dual}_{A,S}(r)$ is defined in Lemma A.1. By Lemma A.1, we have

$$\delta^{dual}_{A,S}(\tilde{r}) < \tilde{\delta}^{dual}_{A,S}(\tilde{r}) \leq \tilde{\delta}^{dual}_{A,S}(1-\tau) < \delta^{dual}_{B,N}(0) \;\Rightarrow\; \delta^{dual}_{A,S}(\tilde{r}) < \delta^{dual}_{B,N}(0).$$

For the second statement,

$$\delta^{dual}_{A,S}(0) = \frac{\frac{1}{2} - \frac{\tau}{2} + \frac{\tau^2}{8}}{\frac{1}{2} - \frac{\tau^2}{4}} > \delta^{dual}_{B,S}(0) = \frac{4-3\tau}{8(1-\frac{\tau}{2}-u)} \iff u < \underbrace{\frac{2 - \frac{9}{2}\tau + 4\tau^2 - \frac{5}{4}\tau^3}{4 - 4\tau + \tau^2}}_{>\frac{\tau}{8} \text{ for } 0<\tau<\frac{1}{2}}. \blacksquare$$

As $\delta^{dual}_{B,k}(r)$ is weakly increasing in r, we know that the minimum δ such that collusion can be possibly sustained under no data sharing is $\delta^{dual}_{B,N}(0) = \frac{6-8\tau}{9(1-\tau-u)}$. Thus, by Lemma A.2., we know that as we go from $r = 0$ to the r such that $\delta^{dual}_{B,S}(r) = \delta^{dual}_{B,N}(0)$, the orders of the critical discount factor for A and B flip under data sharing. Also, because $\delta^{dual}_{A,N}$ ($\delta^{dual}_{B,N}$) is strictly decreasing (increasing) in r, the δ such that they intersect will be lower than $\delta^{dual}_{B,N}(0)$.

Proof of Corollary 2.

Recall that collusion can only be possibly sustained if $\delta > \frac{6-8\tau}{9(1-\tau-u)} > \frac{6-8\tau}{9-9\tau}$. Given our parameter restrictions, we first show

$$\frac{36-9\tau}{(1-\frac{\tau}{2})\delta + \frac{3\tau}{8} - \frac{1}{2}} < \frac{24-8\tau}{(1-\tau)\delta + \frac{8\tau}{9} - \frac{2}{3}} \tag{11}$$

$$\iff 36(1-\tau)\delta + 32\tau - 24 - 9\tau\delta(1-\tau) - 8\tau^2 + 6\tau < 24\delta(1-\frac{\tau}{2}) + 9\tau - 12 - 8\delta\tau(1-\frac{\tau}{2}) - 3\tau^2 + 4\tau$$

$$\iff 12 + 25\delta\tau + 5\tau^2 - 5\delta\tau^2 - 12\delta - 25\tau > 0$$

$$\iff \underbrace{(1-\delta)}_{>0}\underbrace{(12 - 25\tau + 5\tau^2)}_{>0} > 0.$$

Now, it is easy to directly show the two results.

$$\begin{aligned}
\pi^c_{A,S}(r^{dual}_S) > \pi^N_{A,N}(r^{dual}_N) &\iff r^c_S(\frac{1}{2} - \frac{\tau}{8}) - r^c_N(\frac{1}{3} - \frac{\tau}{9}) > \frac{1}{6} - \frac{7\tau}{72} \\
&\iff 12 - \tau + \delta u\left[\frac{24-8\tau}{(1-\tau)\delta + \frac{8\tau}{9} - \frac{2}{3}} - \frac{36-9\tau}{(1-\frac{\tau}{2})\delta + \frac{3\tau}{8} - \frac{1}{2}}\right] > 12 - 7\tau \\
&\iff 6\tau > \delta u\Bigg[\underbrace{\frac{36-9\tau}{(1-\frac{\tau}{2})\delta + \frac{3\tau}{8} - \frac{1}{2}} - \frac{24-8\tau}{(1-\tau)\delta + \frac{8\tau}{9} - \frac{2}{3}}}_{<0 \text{ by } (11)}\Bigg].
\end{aligned}$$

$$\begin{aligned}
\pi^c_{B,S}(r^{dual}_S) < \pi^N_{B,N}(r^{dual}_N) &\iff (1 - r^c_S)(\frac{1}{2} - \frac{\tau}{8}) < (1 - r^c_N)(\frac{1}{3} - \frac{\tau}{9}) \\
&\iff 12 - \tau + \delta u\left[\frac{36-9\tau}{(1-\frac{\tau}{2})\delta + \frac{3\tau}{8} - \frac{1}{2}} - \frac{24-8\tau}{(1-\tau)\delta + \frac{8\tau}{9} - \frac{2}{3}}\right] > 12 - \tau \\
&\iff \frac{36-9\tau}{(1-\frac{\tau}{2})\delta + \frac{3\tau}{8} - \frac{1}{2}} - \frac{24-8\tau}{(1-\tau)\delta + \frac{8\tau}{9} - \frac{2}{3}} > 0.
\end{aligned}$$

B Microfoundation for Assumption in the Stage Game.

We provide a brief microfoundation for the assumption that the platform prefers to keep sellers rather than driving them out of the market. Suppose that there are two competing platforms, A_1 and A_2, that sellers can join and now there is a measure two of consumers. Before making their purchasing decision, consumers now must choose which platform to join. That is, we are assuming that sellers can multi-home and consumers are single-homing. Like the stage game, consumers have inelastic demand for one unit of the good and are differentiated in preference by seller. Now, consumers are also uniformly differentiated in preference for platform (i.e. $l_{A_1} = 0$ and $l_{B_2} = 2$). Specifically, consumer preferences are distributed independently and uniformly on $[0,1] \times [0,2]$. A consumer located at (x, y) who purchases from $i_B \in \{B_1, B_2\}$ on platform $i_A \in \{A_1, A_2\}$ and pays p receives utility

$$U(x, y, p) = 1 - \tau_B |l_{i_B} - x| - \tau_A |l_{i_A} - y| + \beta n_{i_A} - p$$

where $\beta > 0$ represents the indirect network externality for consumers and n_{i_A} is the number of sellers on platform i_A. We keep all remaining assumptions on parameters which will be sufficient to guarantee players compete on the margin for their competitor's consumers. The timing of the stage game is now: (i) Sellers decide which platform(s) to join; (ii) Consumers decide which platform to join; (iii) Platforms set their fees and data sharing policy; (iv) Sellers set their respective prices; (v) Consumers observe prices and make their purchasing decision.

First, observe that if platforms have no incentive to leave sellers with profits below u, then it is a strictly dominant strategy for sellers to join both platforms. In this case, the problem each seller faces is the exact same as the original stage game as there will be measure one of consumers that joins each platform. Next, consider when one platform deviates and fully extracts profit from one seller. Under collusion, this profit gain is $\pi_k^c(r_k^{mkt})$.

Now, we analyze the scenario where one seller joins both platforms and the other only joins one platform. Let B_1 join both and B_2 join A_1 only without loss of generality. Because prices are set after consumers join, seller B_1 will set $p(x) = 1 - \tau_B x$ if under data sharing and set $p = 1 - \tau_B$ if under no data sharing on platform A_2. This gives us

$$\Pi_k = \begin{cases} 1 - \frac{\tau_B}{2} & \text{if } k = S \\ 1 - \tau_B & \text{if } k = N \end{cases}$$

which represents the profit earned by seller B_1 on A_2. Pricing on A_1 will remain the same. Solving for the amount of consumers that join A_1 gives us

$$D_{A_1}(\beta) = \begin{cases} \min\{1 + \frac{\beta}{2\tau_A}, 2\} & \text{if } k = S \\ \frac{1}{2}\min\{1 + \frac{\beta}{2\tau_A}, 2\} + \int_{\frac{1}{2}}^{1} \min\{1 + \frac{\beta + 2\tau_B x - 3\tau_B/2}{2\tau_A}, 2\} dx & \text{if } k = N \end{cases}.$$

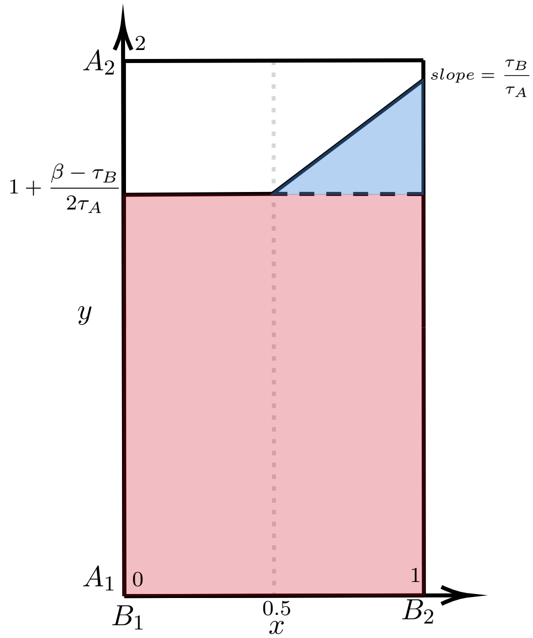

Figure 3: A graphical representation of consumers' platform and seller choice. The red area represents D_{A_1} when $\mathcal{D} = S$. The red + blue area represents D_{A_1} when $\mathcal{D} = N$.

The demand that A_1 receives is derived by solving for the marginal consumer who is indifferent between joining platforms A_1 and A_2 given that they may only purchase from B_1 if they join A_2. Observe that $D_{A_1}(\beta)$ is weakly increasing in β which is quite natural. The larger the seller's indirect network externality is, the more demand a platform would have if it had a larger seller base.

We extend this stage game to the same repeated game and dynamic strategy presented in the paper. To ease exposition, suppose that whenever a platform deviates, the sellers enforce the harshest punishment and never joins the platform again in future periods. Thus, a platform will not deviate and drive one seller out of their platform if

$$\underbrace{\frac{(1 + r_k^{mkt})\pi_k^c(r_k^{mkt})}{1 - r_k^{mkt}}}_{\text{deviation payoff}} + \underbrace{\frac{\delta r_k^{mkt}(2 - D_{A_1}(\beta))\Pi_k}{(1 - \delta)(1 - r_k^{mkt})}}_{\text{off-path payoff}} \leq \underbrace{\frac{2\pi_k^c(r_k^{mkt})r_k^{mkt}}{(1 - r_k^{mkt})(1 - \delta)}}_{\text{original collusive payoff}}$$

$$\iff \beta \geq D_{A_1}^{-1}(2 - \underbrace{\frac{(r_k^{mkt}(1 + \delta) - 1 + \delta)\pi_k^c(r_k^{mkt})}{\delta r_k^{mkt}\Pi_k}}_{\geq 0 \text{ when collusion occurs}}).$$

Therefore, with sufficiently large β, the above condition holds. That is, a platform will opt to make sure both seller profits are above u if the indirect network benefit a seller has on consumers is large enough. This makes sellers sufficiently valuable to platforms as driving a seller out drastically reduces its market demand in future periods. We can conduct similar analyses if the platform deviates with both sellers or if the platform operates in dual mode and we will arrive at a similar result.